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Optimising electromechanical whisker design for contact localisation

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ABSTRACT

Electromechanical whiskers are inexpensive and useful tactile sensor systems for robotic mapping tasks within environments problematic to image and range sensors. This research aims to quantify the effect of the whisker's mechanical properties on its ability to achieve accurate contact localisation. Using mechanical beam theory, a model relating the rate of change of a whisker's strain as a result of the location of a radial contact was derived. A more practical linear approximation of this model was then created with an operating range for which a whisker was theoretically capable of performing contact localisation within 0.1% error. These models facilitated the design of theoretically optimal whiskers. It was discovered that the longer the whisker, and the closer the strain-measuring sensor was placed towards the whisker base, the greater the required taper. For larger whisker widths and faster contact speeds, a smaller taper was required. Additionally, whiskers tapered in 2 dimensions required larger tapers than one-dimensional tapers to achieve optimal performance. The presence of a central hollow region (medulla) did not have a practical effect on the required taper or the performance of the whisker. Experimental results showed that a 200 mm whisker with an optimal taper derived from our model was able to measure the location of contacts with an error less than the target 0.1% anywhere in the distal 58 mm, with a response time of 0.1 s. Tapering electromechanical whiskers to their derived optimal value enhances the accuracy and efficiency of whisker-based robotic mapping.

1. Introduction

The ability for mobile robots to autonomously plan and execute navigation using knowledge of their surroundings gathered from on-board sensors has enabled them to be effective in many modern day affairs including inspection [1], search and rescue [2], entertainment [3] and sanitation [4], among many others. The simultaneous localisation and mapping (SLAM) method is used to informedly perform live navigation based on an environment map generated simultaneously using on-board sensors and robotic mapping [5].

The majority of modern robotic mapping approaches utilise optical rangefinders (e.g., light detection and ranging sensors) or vision sensors (e.g., cameras) [6]. Robotic mapping can be enhanced through the implementation of tactile sensing in environments notoriously precarious for optical sensing, including those containing occluded and concealed objects [7,8], compliant and/or malleable bodies [9], particulate air [10] and reflective and/or transparent surfaces [11]. A major disadvantage of tactile sensors is their comparatively shorter search range than optical or visual sensors due to their reliance on physical interaction with external bodies [12]. However, in contrast to range and imaging sensors, whiskers are not only inexpensive and simple to construct, but require significantly less computational power to operate.

Mammalian facial whiskers (vibrissae) are tactually-sensitive hairs that provide tactile exteroception in almost all mammals [13]. Many small quadrupedal mammals use the information from vibrissae to guide locomotion [14]. Rats commonly feature arrays of between 30 to 35 macrovibrissae of length ranging between 4 mm and 50 mm on either side of the snout [15,16] (2%–23% of the mean length of the brown rat, 217 mm [17]). Rat whiskers can be approximated as truncated cones, and contain a low density region known as the medulla within the proximal portion [18]. The tapered profile enables whiskers to slip past objects once a critical bending angle has been realised [19], enhancing mobility through cluttered environments, whilst the medulla reduces sagging due to gravity [20]. The length and diameter of both the cuticle and the medulla vary based on the location of the whisker on the mystacial pad, and has been quantified in [21]. The formation of whiskers on the mystacial pad forms 2 search volumes on either side of the snout with a shape resembling sections of spheres centred about the rat's eyes [22]. Rats will actively sweep (whisk) their whiskers in front of them in the azimuth direction to induce light contacts with external objects [23,24], further expanding the whisker search

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volume. Behavioural studies indicate that rats will whisk in a manner to purposefully make contact at radial locations near to, and at, the whisker tip [24].

Two different types of mechanoreceptive cells are located in the whisker follicle, allowing for the detection of both the magnitude and rate of change of the whisker deflection [25,26]. Neurological processing of the mechanoreceptor signals, proprioceptive knowledge of whisker locations on the snout, and the timing of the contact during the whisk cycle allows rats to determine the spatial location of a contact [27,28]. Through this, rats can recognise both the shape [15] and orientation [29] of objects.

Contacts made on an electromechanical whisker can be localised by deciphering the mechanical response of the whisker measured by a variety of sensors. Selection of the sensor device(s) used, the whisker shaft material, and the manufacturing process are key design choices for electromechanical whiskers, and are often intertwined. Following the foundational concept that animal nervous systems are more adept at measuring the rate of change of signals rather than their absolute magnitude, the rate of change of the whisker strain [30] or the bending moment [31] collected by strain gauges can be converted to a contact location for whiskers arranged as cantilever beams. In the case of the whisker strain, this holds true as long as the whisker undergoes small deflections, e.g., less than 10 degrees [30]. Similar outcomes can be achieved through measurement of the pressure on a load plate perpendicular to the whisker shaft using barometers [32,33] and piezoresistors [34]. Measurement of the degree of deformation throughout the whisker shaft can be used, achievable by measuring the wavelength shift due to bending of optical fibre [35]. For a hinged whisker, the displacement of the whisker base in multiple dimensions can be used to convert whisker shaft deformation into a contact location using Hall effect sensors [36,37]. A hinged whisker is seemingly more complex to develop than a cantilever beam due to the requirement for an automatic return system, which can be especially challenging to achieve at small scales. Many other designs utilising different sensing modalities also exist (see [38] for a recent review covering electromechanical whisker designs). In all of these cases, the signal magnitude varies non-linearly with the contact location.

The whisker shaft can be manufactured simply as beams from plastic rods [32,34] or metal wires [31,37]. Curvature can be achieved through heat-forming [37]. 3D printing [33,39] and moulding [30] allows for precise manufacturing of whiskers with various shapes. Thermoplastics often used in 3D printing, such as acrylonitrile butadiene styrene (ABS), possess favourable mechanical properties which allow whiskers to deflect elastically at large angles (up to at least 50 degrees) without permanent (plastic) deformation [40], which can be useful for operation within confined spaces. Fibre drawing of thermoplastics can achieve desired tapering of whiskers at smaller scales than conventional fused deposition modelling [40]. The whisker can be formed entirely from the sensor system itself, as in the case for optical fibre whiskers [35], and embedding such sensors within a material can control its deformation properties whilst protecting the sensor from the external environment [41,42].

Maps can be generated using electromechanical whisker sensing through the collection of localised contact locations in space over the duration of a navigation mission. Contour fitting allows for the reconstruction of object shapes using localised whisker contact points [8,31,35]. Modelling of a robot's searched environment can be achieved through population of an occupancy map using recorded whisker contact points, achievable using SLAM techniques [43–45]. Texture sensing is another heavily explored application with electromechanical whiskers, allowing for discrimination of objects based on their surface smoothness [46–48].

Both the bending moment and the strain along the whisker increases in direct proportion to the whisker's base diameter and the contact speed [30,31]. As the width of the whisker increases, however, so will its rigidity [49]. For rats, the conical whisker taper is an important functional feature as this allows objects to conveniently slip past the whisker without inducing large, potentially damaging bending angles, enabling agile mobility within their typically confined and cluttered habitats [19,50]. Untapered whiskers will also frequently become stuck inside the grooves of textured surfaces, whereas tapered whiskers, especially conical, will undergo characteristic stick–slip behaviour, indicating a greater applicability of tapered whiskers to feature sensing [50]. For electromechanical whiskers, the three-dimensional (3D) contact location can be uniquely calculated using the combination of the axial force, the bending moment magnitude, and the direction of the bending moment measured at the whisker base, as long as the whisker is tapered [51]. 3D contact localisation has been found to be more accurate for tapered whiskers than untapered whiskers when using these same mechanical measurements [40]. A "sweet spot" between the location of object contact and the contact speed for which discrimination of contacts are more easily achieved is suggested to exist [52]. However, this balance has not been identified across the various parameters that make up a whisker system.

Aside from these observations, the impact of an electromechanical whisker's geometry and other design/functional parameters on its capability to localise contacts has, to the best of our knowledge, yet to be quantified. The goal of this research was to universally quantify the optimal balance between whisker length, thickness, taper, and whisk speed for achieving accurate contact localisation. The impact of the presence of a medulla was also explored. This was with specific interest towards whisker systems that can achieve accurate localisation as efficiently as possible, requiring minimal computing resources for signal processing. This knowledge will enable roboticists to efficiently design electromechanical whisker systems that can attain optimal contact localisation results, improving robotic mapping outcomes.

2. Materials and methods

2.1. Whisker parameters and nomenclature

Figs. 1(a) and 1(b) show, respectively, the front and top view of a schematic electromechanical whisker of length l_w, base width w_b, and tip width w_t undergoing contact by an object at distance l_c from the whisker base with contact speed v_c. Our approach treated whiskers as straight, homogeneous, flexible cantilever beams due to the simplicity in manufacture of a cantilever whisker compared to a hinged whisker. In a similar fashion, the whisker strain, measurable through a single sensor element such as a strain gauge placed at a distance l_s from the whisker base, was used as the sensory input for contact localisation as this requires straightforward assembly with simple circuitry and signal processing as opposed to other methods.

The whisker's taper magnitude (T) was defined as the ratio of the cross-sectional area of the outer shape of the whisker base (A_b) to the cross-sectional area of the outer shape of the whisker tip (A_t), disregarding any holes present in either of the cross-sections, as per Eq. (1).

$$T = \frac{A_b}{A_t} \tag{1}$$

We explored 2 categories of whisker taper: (1) tapered in a single dimension (T_{1D}); and (2) tapered in 2 dimensions (T_{2D}). A T_{1D} whisker was one whose height remains constant but whose width decreases along the whisker length (represented as a dashed line in Fig. 1). A T_{2D} whisker was one in which both the width and the height decreased uniformly along the whisker length (represented as a solid line in Fig. 1), e.g., a cone. Whiskers with no taper (i.e., $T = 1$), e.g., a cylinder, were classified as untapered (T_{0D}). This research only considered whiskers with a linear taper with the base's external cross-section having regular shape (i.e., a regular polygon or a circle). Both T_{1D} and T_{2D} taper types were explored as we hypothesised that they would each provide distinct advantages in different situations. T_{1D} whiskers, being more

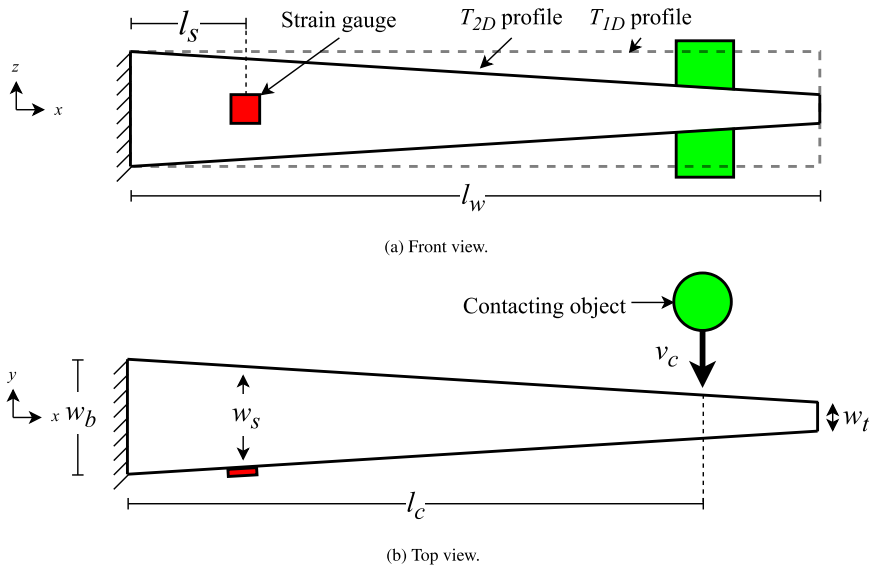


Fig. 1. Schematic of a T_{1D} (dashed profile) and T_{2D} (solid profile) whisker of length l_w, base width w_b and tip width w_t, undergoing contact at length l_c from the whisker base at contact speed v_c. l_s represents the distance between the whisker base and the centre of the sensing element, with w_s being the width of the whisker at this location. (**a**) Front view. Only T_{2D} whisker tapers in height. (**b**) Top view. Both T_{1D} and T_{2D} whiskers taper in width.

flexible in the direction of the contact (y axis) than in the perpendicular direction (z axis), would be expected to be more suitable to applications where only unidirectional contacts were of interest, as deflection in the perpendicular direction would be more restricted. T_{2D} whiskers, being flexible in both the y and z directions, were expected to be more suitable for measuring contacts occurring in both planes.

2.2. Derivation of whisker strain relationship to contact location

Under the assumption that the contacting object is rigid and applies a point load to the whisker during contact, the Euler–Bernoulli beam equation [49] can be used to derive a numerical relationship between the whisker strain and the contact location. For a small deflection in the y direction (s) at the contact location (l_c) arising from a point load (F) (e.g., a rigid body contact), the curvature (κ_s) of a cantilever beam (whisker) measured at a point on the whisker face along the x axis (l_s) is given as:

$$\kappa_s \approx \frac{d^2y}{dl_s^{\,2}} \approx \frac{M_s}{EI_s} = \frac{F(l_c - l_s)}{EI_s}, \tag{2}$$

where M_s is the moment at point l_s, E is the Young's modulus of the whisker, and I_s is the second moment of area at point l_s in the direction perpendicular to the contact. Study of the geometry of tapered beams shows that the width of the whisker at the sensor location (w_s) is related to the whisker base width (w_b), total length (l_w), and the distance of the sensor from the base (l_s) by:

$$w_{s_{1D}} = w_b\left(1 + \frac{l_s}{l_w}(1/T - 1)\right) \tag{3a}$$

$$w_{s_{2D}} = w_b\left(1 + \frac{l_s}{l_w}\left(\sqrt{1/T} - 1\right)\right) \tag{3b}$$

where $w_{s_{1D}}$ is the whisker width at the sensor location for a T_{1D} whisker, and $w_{s_{2D}}$ is that for a T_{2D} whisker. I_s can therefore be written as the product of the second moment of area at the whisker base (I_b) and the scale of diversion of the second moment of area at the sensor location from that at the whisker base (∇I), as in Eq. (4):

$$I_s = I_b \nabla I \tag{4}$$

$$\nabla I_{T_{1D}} = \left(1 + \frac{l_s}{l_w}(1/T - 1)\right)^3 \tag{4a}$$

$$\nabla I_{T_{2D}} = \left(1 + \frac{l_s}{l_w}(\sqrt{1/T} - 1)\right)^4 \tag{4b}$$

where $\nabla I_{T_{1D}}$ is the scale of diversion of the second moment of area at the sensor location from the whisker base for a T_{1D} whisker, and $\nabla I_{T_{2D}}$ is that for a T_{2D} whisker. This relationship holds true for both solid and hollow whiskers, as long as the taper magnitude of the hollow geometry matches the taper magnitude of the whisker's shaft.

Substituting Eq. (4) into Eq. (2), taking the definite integral from points 0 to l_s twice to solve for the whisker displacement, and solving for the contact location, an equation for the deflection at the contact location (s_T) can be obtained for both taper classes:

$$s_{T_{1D}}(l_c) = \frac{Fl_w(a_1{}^2 l_c{}^2 + 2l_w{}^2(\ln(a_1 l_c + l_w) - \ln(l_w)) - 2a_1 l_c l_w)}{2a_1{}^3 EI_b} \tag{5a}$$

$$s_{T_{2D}}(l_c) = \frac{Fl_c{}^2 l_w(2l_c{}^2 a_2 + 2l_c l_w)}{6EI_b(l_c a_2 + l_w)^2} \tag{5b}$$

where $a_1 = 1/T - 1$ and $a_2 = \sqrt{1/T} - 1$. The whisker strain at the sensor location (ϵ_s) is related to the moment at l_s (M_s), section modulus at the sensor location (Z_s), and Young's modulus of the whisker (E) by Eq. (6):

$$\epsilon_s = \frac{M_s}{Z_s E} = \frac{F(l_c - l_s)}{Z_s E} \tag{6}$$

Rearranging Eq. (6) for F and substituting into Eq. (5), the displacement at the contact location can be expressed in terms of the whisker strain measured at the sensor location:

$$s_{T_{1D}}(l_c) = \epsilon_s \times \frac{l_w Z_s(2l_w{}^2 \ln(a_1 l_c + l_w) + a_1 l_c(a_1 l_c - 2l_w) - 2l_w{}^2 \ln(l_w))}{2a_1{}^3 I_b(l_c - l_s)} \tag{7a}$$

$$s_{T_{2D}}(l_c) = \epsilon_s \times \frac{l_c{}^3 l_w Z_s}{3I_b(l_c - l_s)(l_c a_2 + l_w)} \tag{7b}$$

Solving Eqs. (7a) and (7b) for ϵ_s and taking the time derivative on both sides, the relationship between the rate of change of strain ($\dot{\epsilon}_s$) and the whisker contact location is:

$$\dot{\epsilon}_{s_{T_{1D}}}(l_c) = v_c \times \frac{2a_1{}^3 I_b(l_c - l_s)}{l_w Z_s(2l_w{}^2 \ln(a_1 l_c + l_w) + a_1 l_c(a_1 l_c - 2l_w) - 2l_w{}^2 \ln(l_w))} \tag{8a}$$

$$\dot{\epsilon}_{s_{T_{2D}}}(l_c) = v_c \times \frac{3I_b(l_c - l_s)(l_c a_2 + l_w)}{l_c{}^3 l_w Z_s} \tag{8b}$$

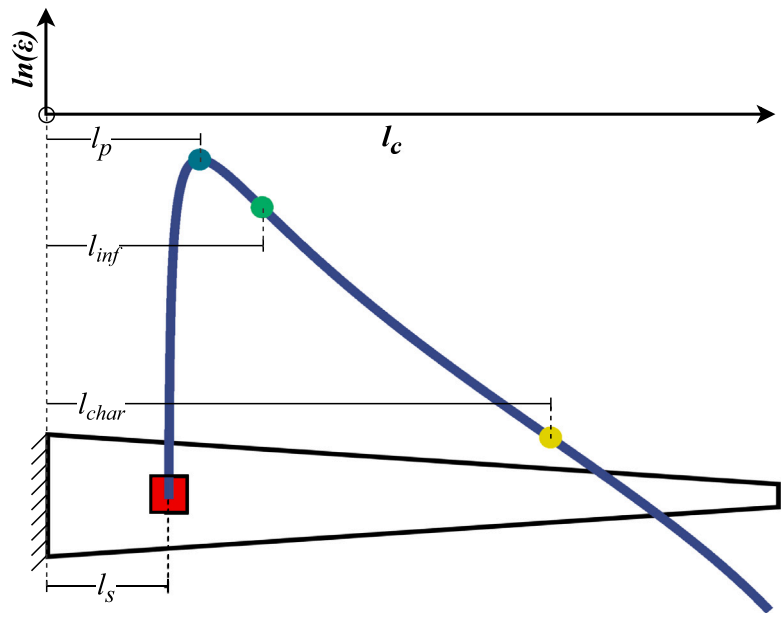


Fig. 2. General shape of the logarithm of the strain rate of change across contact location (l_c) for an arbitrary whisker design (here shown T_{2D} taper, refer to Section 2.1) overlaid over the whisker (blue solid non-linear line). The contact location that produces the peak strain rate of change was represented by l_p. The contact locations of the first and second (characteristic) inflection points were represented by l_{inf} and l_{char}, respectively. The locations of these points can be found by setting the first (l_p) and second (l_{inf} and l_{char}) derivatives of Eq. (8b) to zero. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

If the contact velocity (v_c) is known, which is achievable by performing controlled motorised whisks, solving Eqs. (8a) and (8b) for l_c can therefore be used to deduce the contact location from the rate of change of the whisker strain at location l_s for small whisker deflections. It is important to note that Eq. (8a) is invalid for a T_{0D} whisker as it means $a_1 = 0$, which results in $\dot{\epsilon}_{s_{T_{1D}}} = 0$, which is untrue. Therefore Eq. (8b) must be used for untapered whiskers. It is also important to note that the shape of the whisker's cross-section does not feature in the equations. Therefore, the rate of change of the strain is independent of the cross-sectional shape for whiskers within the same taper class which have a uniform taper. Additionally, Eqs. (8a) and (8b) show that $\dot{\epsilon}_s$ does not depend on the whisker's material (i.e., the Young's modulus). This is as the equations assume that the whisker's (or inversely the object's) translational velocity remains constant during the contact, and that the bending of the whisker during the contact does not contribute to a reduction in this velocity. This is practically suitable assuming that the contacting object is rigid and the whisker is flexible enough to not undergo plastic deformation from the applied displacement.

2.3. Linearising the model

Fig. 2 shows the general shape of the logarithm of the strain rate of change across contact location curve for a whisker of arbitrary length, taper, width, and sensor location.

The shape of the logarithm of the strain rate of change across contact location reveals three fundamental points along the curve: Location of the peak strain rate of change l_p; location of the first inflection point l_{inf}; and location of the second inflection point l_{char}. Firstly, the strain rate of change rapidly increases for contact locations further along the whisker length than the sensor location before reaching a peak at location l_p, for which the strain rate of change decreases beyond. The contact location producing this peak indicates the lower limit of the total operating range of the whisker, as contacts towards the whisker base from this point will produce non-unique strain rate of change with contacts towards the whisker tip, making the contact location ambiguous.

The first inflection point, l_{inf}, exists a short distance following the peak. However, its value was not important to our method. The second inflection point, l_{char}, exists towards the whisker tip. This second inflection point was of importance for obtaining a linear approximation of the whisker's strain rate of change across contact location, and will henceforth be referred to as the characteristic inflection point. Deriving a linear approximation centred at this point will create a linear operating range towards the whisker tip, which was desirable for mapping purposes.

The linearised model of the logarithm of the strain rate of change across contact location was calculated for T_{0D}, T_{1D} and T_{2D} whiskers as the first order Taylor series centred about the whisker's characteristic inflection point. The operating range for a whisker's linear model was defined as the region of the whisker for which the percent error between the strain rate of change curve and the first order Taylor series approximation was within 0.1%. This maximum error was selected to promote high accuracy within the entirety of the linear operating range.

A whisker's optimal taper (T^*) was defined as that which possessed the widest linear operating range that included the whisker tip, thus maximising the operating range. This range also enables mimicry of rat whisker placement behaviour which is biased towards contacts near and including the tip [24]. T^* was calculated by finding the smallest whisker taper that possessed a linear operating range whose upper limit was located exactly at the whisker tip. Generalised mathematical models describing T^*, the corresponding size of the linear operating range (W^*) and location of the characteristic inflection point (l^*_{char}), were developed by calculating these values for whiskers of varying length, sensor location, base width and contact speed for both T_{1D} and T_{2D} whiskers. The sensor location was limited to within 1/8th of the whisker length from the base as sensors placed within this region would have reduced potential for collision with objects during robotic mapping tasks.

The effect of deviating from the optimal taper on whisker-based robotic mapping was observed by plotting the linear operating size range against the proportion of whisker taper against the optimal whisker taper. This was performed for both T_{1D} and T_{2D} whiskers across varying sensor location, whisker length, base width and contact speed.

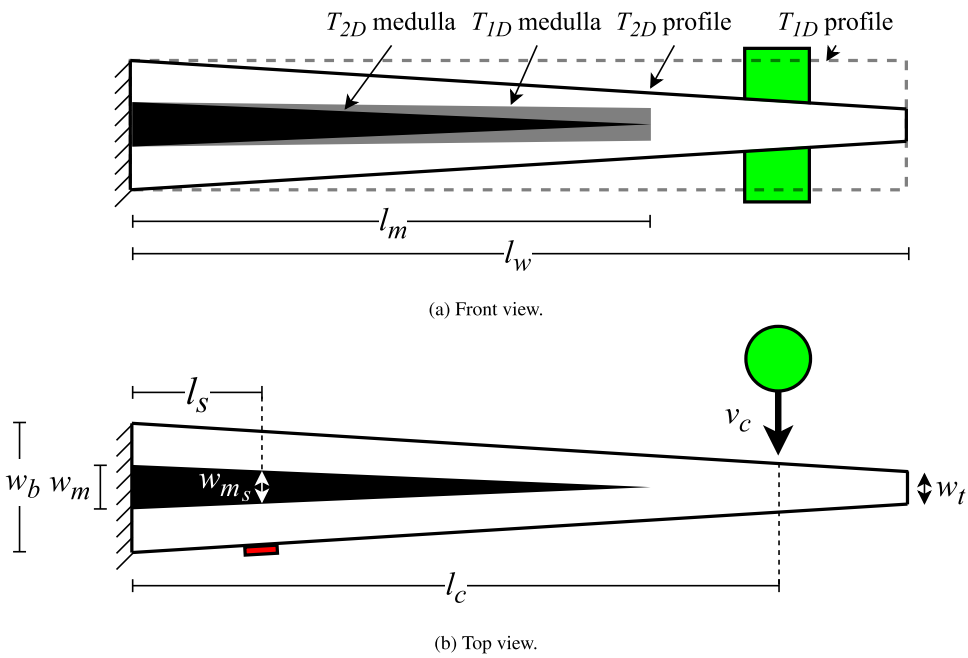


Fig. 3. Schematic of the central cross-section of whiskers with a biologically mimetic medulla (shaded area) with base width w_m and length l_m. The darker shaded region shows the shape of the medulla for T_{2D} whiskers, and the lighter shaded region shows the shape of the medulla for T_{1D} whiskers. The width of the medulla at the sensor location is denoted w_{m_s}. **(a)** Front view. The medulla for T_{2D} whiskers tapers to a point, but the medulla for T_{1D} whiskers tapers to an edge. **(b)** Top view. The medulla for both T_{1D} and T_{2D} whiskers taper in width.

2.4. Exploring the medulla

The effect of a biologically mimetic medulla on whisker performance was explored mathematically. Here, whiskers were now modelled to contain a medulla as a tapered, hollowed-out region centred about the central axis of the whisker. The medulla's cross-section had the same shape of the cross-section of the whisker shaft, with the widest point at the base, and tapered linearly with the same taper dimensionality as the whisker shaft (i.e., a 1D taper for T_{1D} whiskers and a 2D taper for T_{2D} whiskers) up to an edge (for T_{1D} whiskers) or a point (for T_{2D} whiskers), at a distance l_m from the whisker base. Fig. 3 shows the central cross-section of a whisker containing a medulla, showing these parameters. The base width of the medulla (w_m), and the length of the medulla (l_m), were fixed for any given whisker based on Eqs. (9a) and (9b) respectively, selected based on the average medulla proportions for the column of caudal-most whiskers on the rat snout (i.e., the longest whiskers) [21]. This resulted in the width of the medulla at the sensor location (w_{m_s}) being related to the whisker base width, length and sensor location by Eq. (9c).

$$w_m = 0.324 w_b \tag{9a}$$

$$l_m = 0.786 l_w \tag{9b}$$

$$w_{m_s} = w_b \left(0.324 - 0.415 \frac{l_s}{l_w}\right) \tag{9c}$$

The scale of diversion of the second moment of area at the sensor location from that at the whisker base for whiskers not containing a medulla (∇I) was no longer equal to the relationships of Eqs. (4a) and (4b) for whiskers containing a medulla where the taper magnitude of the whisker was such that the slope of the whisker shaft did not match the slope of the medulla. The scale of diversion of the second moment of area at the sensor location from that at the whisker base was derived for T_{1D} and T_{2D} whiskers containing medullae (∇I_m), and the ratio of (∇I_m) to ∇I for whiskers of the same taper class was taken to quantify the impact of the addition of the medulla on this parameter (see Appendix A for the algebraic derivations of these). This revealed that ∇I_m was practically equivalent to ∇I, and therefore Eqs. (8a) and (8b) were deemed appropriate for calculating the strain rate of change of the whisker due to a contact for whiskers possessing a medulla.

The optimal taper, operating range, and characteristic inflection point was modelled for whiskers possessing a medulla using the same method as in Section 2.3. The effect of the medulla on these parameters was quantified by calculating the root mean square error (RMSE) between the models for whiskers with a medulla to those without medullae.

2.5. Simulations

Transient structural analyses were performed in Ansys Mechanical 2021 R2 to verify the correctness of the derived strain-contact location relationship. Whiskers were fixed at the base using a fixed body-ground joint, and an 8 mm diameter steel cylinder was driven radially into the whiskers with constant speed at various contact locations (refer to Section 2.1 and Fig. 1 for more details). A cylinder was used to produce a constant translational deflection at a point on the whisker whilst allowing the whisker to slip smoothly past it as it deforms. Note that this causes the contact location to slide towards the whisker tip as the whisker deforms. As we were only interested in the strain relationship over small deflections, the effect of this slippage was negligible. The strain along the whisker length on the side opposite to the contact was measured across time, which was then used to calculate the strain rate of change at potential sensor locations.

Two variations in the whisker cross-sectional shape were studied for the three taper classes (T_{0D}, T_{1D} and T_{2D}) to check for consistency in results for varying whisker profiles. For T_{0D} whiskers, both cylindrical and square prisms were studied. For T_{1D} whiskers, whiskers were designed as a square base tapering to a rectangular tip and a circular base tapering to an ellipsoidal tip (both with constant heights). For T_{2D} whiskers, truncated square pyramids and truncated cones were studied. Three different whisker base widths were simulated to observe the effect of the whisker width: 4.5 mm, 6 mm and 8 mm. These widths were selected as whiskers could be reliably manufactured at these widths for comparison between simulation and hardware.

The resulting strain rate of change data from the simulation results were plotted against the theoretical model to observe the closeness

Table 1
Materials and their respective Young's modulus (as quantified in Ansys) used in the simulations to determine the effect of whisker material on the strain rate of change.

Material	ABS	PLA	Epoxy resin	Beech wood	Aluminium alloy	Structural steel
Young's Modulus [GPa]	2.41	2.94	3.78	12.93	71	200

of the simulation results to theory. The coefficient of determination (R^2) and RMSE were calculated for each whisker taper to quantify the alignment of the simulation results to the theory.

Simulations were additionally performed to determine whether there was an impact on the strain rate of change relationship with the contact location due to the whisker's material. The strain at the sensor location was recorded for whiskers of equivalent dimensions but made from different materials. The materials tested, and their Young's modulus (recorded from Ansys), can be found in Table 1. The resulting strain rate of change during the initial contact period (i.e., small deflection period) was compared for equivalently shaped whiskers of different materials for contacts made at the same location.

2.6. Hardware experiments

Experiments were performed with a manufactured ABS whisker with optimal T_{2D} taper to observe the performance of a theoretically optimal whisker in performing contact localisation using the developed linear approximation method. The whisker was manufactured as a truncated square pyramid with w_b=4.5 mm and l_w=200 mm through fused deposition modelling 3D printing. This method allowed for inexpensive and repeatable whisker production. An alternative approach of printing whiskers from inexpensive resin was found to limit their use in confined spaces due to their tendency to snap at smaller angles than ABS whiskers, and was therefore not used. While specialist resins may have been available, their cost would have been prohibitive. Derivation of the taper magnitude used for this whisker can be found in Section 3.3. A truncated square pyramid was used to promote deflection in the x-y plane and hinder deflection in the x-z plane from a contact in the y direction (see Fig. 1(b)) whilst resembling the T_{2D} geometry seen in rat whiskers. The whisker length was selected such that the whisker could be useful for robotic mapping when aboard a small mobile robot, where a maximum detection range of 200 mm would be sufficient for object avoidance and to cover potential dead zones of LiDARs. The base width selected was the minimum practical base width that allowed for adhesion of a strain gauge (Tokyo Measuring Instruments Lab model FLAB-5-11-1LJC-F) at a sensor location of $l_s = 20$ mm. This was the lowest practical sensor location within the targeted bottom 1/8th of the whisker length due to the strain gauge dimensions and wire routing considerations. The strain gauge was wired to an arm of a balanced quarter Wheatstone bridge whose output was amplified with an instrumentation amplifier. The amplified output was connected to a 12-bit analogue to digital converter (ADC) pin of a Teensy 4.1 to measure the resulting resistance change from increasing whisker strain at a sampling rate of 500 Hz (sampling period 2 ms).

Fig. 4 illustrates the experimental setup. The whisker was mounted to a custom steel clamp that clasped a 40 mm extension from the whisker base, forming a cantilever beam with free end of length l_w. A motorised linear rail was positioned perpendicular to the whisker which drove a carriage holding a shanked M8 bolt travelling radially to the whisker. The bolt was positioned such that only the smooth, shanked area would make contact with the whisker. The rail was driven at a maximum linear speed of 16 mm/s using a stepper motor. The experimental setup was situated upon an air-stabilised optical fixture table (Newport SmartTable) with the motorised rail fixed in position. The whisker clamp was able to be moved closer to and further from the rail to allow for variable positioning of the contact location.

The measured ADC value from the output of the amplified Wheatstone bridge was sent to a MATLAB script that used to save the transient response of the ADC during the contact. The rate of change of the ADC value was determined by calculating the temporal derivative. This signal was noisy, so a low-pass filter was used to smooth the result. The filtering used was the fastest first-order infinite impulse response (IIR) time constant that resulted in the difference between average recordings of baseline (no contact) and tip contact being two standard deviations ($\tau = 0.046$ s). This approach was chosen as it could be implemented as a hardware filter if desired. It also had the minimum impact on the dynamics of the system while achieving a functionally useful level of detection accuracy at the whisker tip. An additional top-hat analysis window was used to analyse the average transient response of the ADC during contact. The analysis window was measured for a window of fifty samples (0.1 s period) beginning once the filtered ADC derivative transient response reached steady state.

To convert from ADC rate of change to strain rate of change, a scalar equal to the theoretical strain rate of change at the whisker's characteristic inflection point divided by the average ADC rate of change measured at the characteristic inflection point was used. This single scaling factor was selected as it represented a practical and efficient method of calibrating the hardware sensor rather than taking multiple points along the whisker length.

The strain rate of change was measured for contacts made at the tip down to 40 millimetres above the whisker base at 2 mm spacings. This was limited by the experimental apparatus. Following the single-point calibration process, the R^2 and RMSE of the experimental results compared to the theoretical model were calculated for the whisker to determine the accuracy of the whisker.

The whisker's measured strain rate of change values were converted to an estimate of the contact location by passing them through the inverse of the whisker's theoretical linear approximation. Values exceeding the whisker length were clamped to be equal to the whisker length. The error between the estimated contact location and actual contact location was quantified across the whisker length as a measure of the difference between the measured contact location and the real contact location to determine the accuracy of the linearisation approach across the whisker length.

3. Results

3.1. Optimal whisker taper

The magnitude of the optimal whisker taper (T^*) was found to be dependent on the whisker's length (l_w), base width (w_b), sensor location (l_s) and contact speed (v_c). Additionally, it differed for T_{1D} and T_{2D} whiskers. The relationship between these parameters, and the magnitude of the optimal whisker taper, was modelled by Eqs. (10a) and (10b) for T_{1D} and T_{2D} whiskers respectively:

$$T^*_{1D} \approx l_w^{0.029}\left(3.966 - 2.348\left(\frac{l_s}{l_w}\right)\right) - \ln(w_b v_c)\left(0.138 - 0.0108\ln(l_w) - 0.0692\left(\frac{l_s}{l_w}\right)\right), \left\{0 \le \frac{l_s}{l_w} \le 0.125\right\} \tag{10a}$$

$$T^*_{2D} \approx l_w^{0.039}\left(8.788 - 8.115\left(\frac{l_s}{l_w}\right)\right) - \ln(w_b v_c)\left(0.401 - 0.028\ln(l_w) - 0.275\left(\frac{l_s}{l_w}\right)\right), \left\{0 \le \frac{l_s}{l_w} \le 0.125\right\}, \tag{10b}$$

where the length from the base to the sensor, the total whisker length and the width of the base are in mm and the contact velocity is in mm/s. Refer to Figs. B.1 and B.2 in Appendix B for plots of the

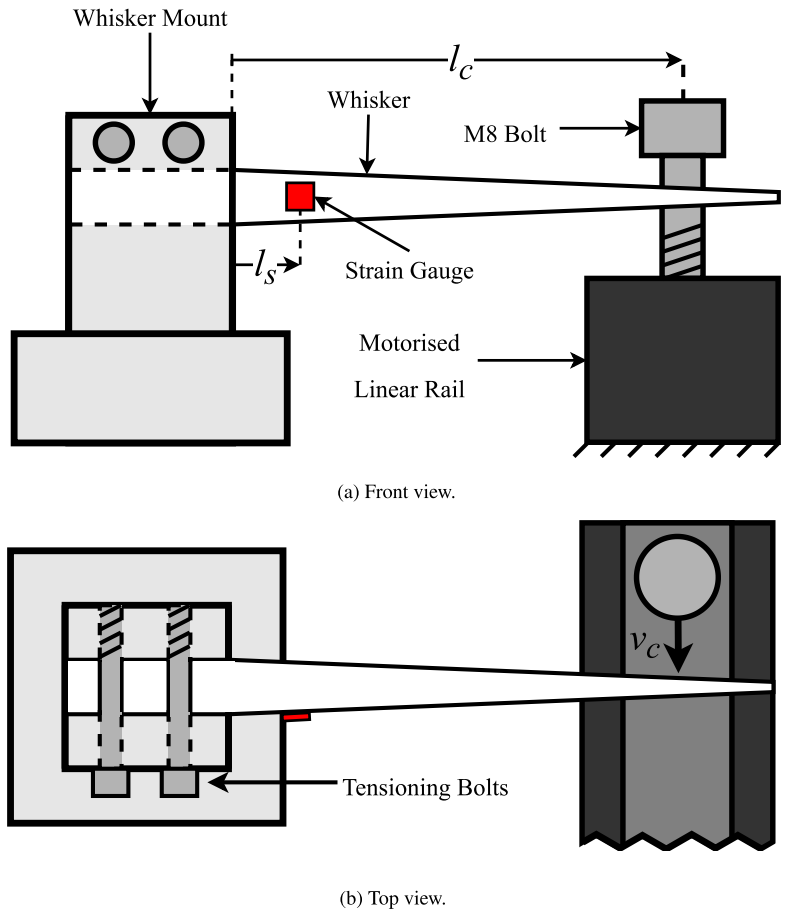


Fig. 4. Diagram of the experimental setup. A straight extension at the whisker's rear was fastened inside a steel mount using two bolts to create an interference fit, enabling the free end to act as a cantilever beam with the longitudinal axis parallel with the ground. A shanked M8 bolt was driven radially into the whisker at position l_c along the whisker length (variable by adjustment of the mount's location) with contact speed v_c by a motorised linear rail to induce a small, constant deflection. (**a**) Side view. (**b**) Top view.

calculated optimal taper (from theory) and the approximated optimal taper (from Eqs. (10a) and (10b)) across the four parameters.

For realistic scenarios (e.g., whisker lengths below 10s of metres, base widths in the order of mm and contact speeds in the order of mm/s to m/s), longer whiskers required marginally larger tapers. Alternatively, a smaller taper was required if the sensor location was placed further from the base of the whisker. A smaller whisker taper was also required as the whisker's momentum (which scales proportionally with w_b and v_c) was increased. T_{2D} whiskers required larger tapers than T_{1D} for equivalent l_w, l_s, w_b, and v_c. However, the rate of change of the whisker's width along the whisker length was larger for T_{1D} whiskers (i.e., the T_{1D} whisker had a smaller tip width than the equivalent T_{2D} whisker). There was no practical difference in the optimal taper between whiskers with and without medullae, with the RMSE between T^* of whiskers with a medulla to those without a medulla being 0.005 for T_{1D} whiskers and 0.008 for T_{2D} whiskers. The generalised models deviated minimally from the calculated optimal tapers, possessing an inherent average error and standard deviation of 0.002±0.015 and −0.009±0.038 for T_{1D} and T_{2D} whiskers, respectively.

Eq. (11) describes the size of the linear operating range for optimally tapered whiskers (W^*) in proportion to the total whisker length l_w:

$$\frac{W^*}{l_w} \approx l_w^{\,0.053}\left(0.229 + c_W + 0.179\left(\frac{l_s}{l_w}\right)\right) - \ln(w_b v_c)\left(0.025 - 0.00268\ln(l_w) + 0.012\left(\frac{l_s}{l_w}\right)\right), \quad \left\{0 \le \frac{l_s}{l_w} \le 0.125\right\}, \tag{11}$$

where l_s, l_w and w_b are in mm, v_c is in mm/s and the operating range size constant (c_W) was −0.0056 for a T_{1D} whisker and 0.0056 for T_{2D} whiskers. Refer to Figs. C.1 and C.2 in Appendix C for plots of the calculated linear operating range size (from theory) and the approximated linear operating size (from Eq. (11)) across the four dependent parameters.

The linear operating range as a ratio of the whisker length for optimal tapers was marginally larger for longer whiskers, and increased as the sensor location moved further from the base. However, an increase in the whisker's momentum lead to a decrease in the size of the operating range. T_{2D} whiskers possessed wider operating ranges than T_{1D} whiskers of equivalent length, sensor location and momentum factor based on the size constant. There was no practical difference in operating range between whiskers with and without a medulla. The average error and standard deviation between the calculated and estimated size was 0.008±0.003 for T_{1D} whiskers and −0.002±0.005 for T_{2D} whiskers.

The location of the characteristic inflection point for optimally tapered whiskers (l^*_{char}) as a fraction of whisker length was modelled in Eq. (12):

$$\frac{l^*_{char}}{l_w} \approx l_w^{\,-0.00845}\left(0.887 + c_{l_{char}} - 0.12\left(\frac{l_s}{l_w}\right)\right) + \ln(w_b v_c)\left(0.0115 - 0.00123\ln(l_w) + 0.00511\left(\frac{l_s}{l_w}\right)\right), \quad \left\{0 \le \frac{l_s}{l_w} \le 0.125\right\}, \tag{12}$$

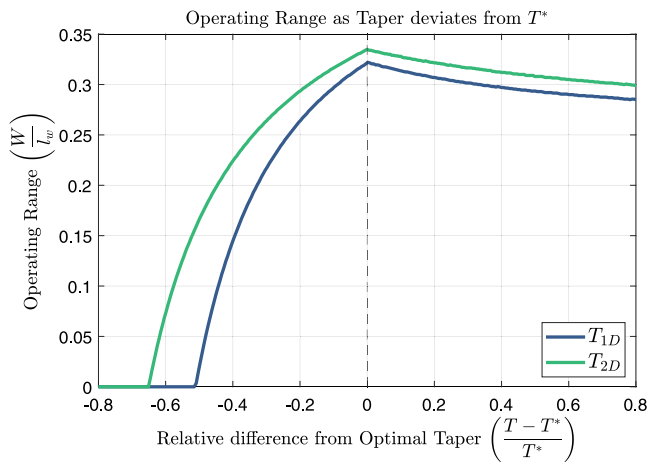


Fig. 5. Operating range change for varying whisker taper for a T_{1D} and T_{2D} whisker with equivalent length $l_w = 200$ mm, sensor location $l_s = 20$ mm ($0.1l_w$) and product of base width and contact speed (i.e., momentum) $w_b \times v_c = 1$ mm^2/s.

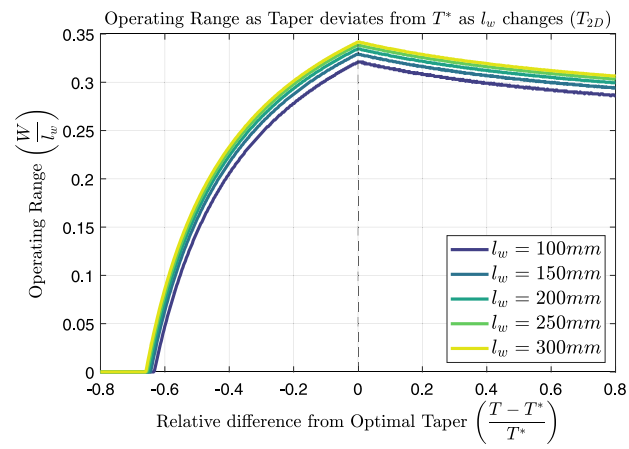
where l_s, l_w and w_b are in mm, v_c is in mm/s per second and the constant of inflection ($c_{l_{char}}$) was 0.0043 for T_{1D} whiskers and −0.0043 for T_{2D} whiskers. As the characteristic inflection point was the point in which the Taylor series approximation was centred at, this model described the ideal practical contact location for the most accuracy (zero error). Refer to Figs. D.1 and D.2 in Appendix D for plots of the calculated linear operating range size (from theory) and the approximated linear operating size (from Eq. (12)) across the four dependent parameters.

Longer whiskers, and those with the sensor placed further from the base, possessed a point of most accuracy closer towards the base. However, as with the size of the operating range, the effect of the whisker length was marginal. Increasing the whisker's momentum caused the point of most accuracy to move closer to the whisker tip. As the whisker's centre of mass moved relatively closer to the base, so did the inflection point. $c_{l_{char}}$ indicated that the point of most accuracy was closer to the base for T_{2D} whiskers than T_{1D} whiskers. l^*_{char} was practically unchanged with the presence of a medulla.

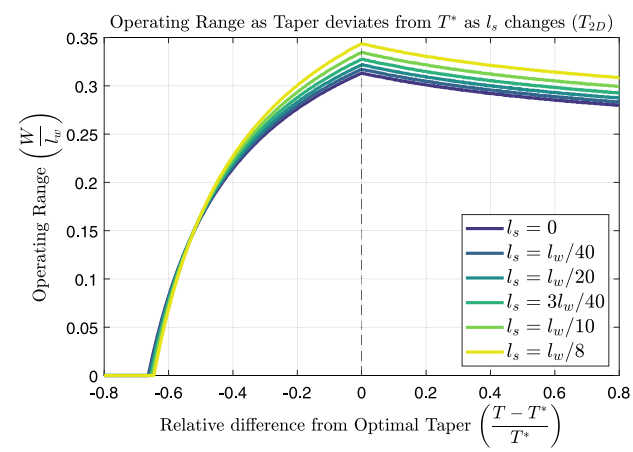
3.2. Effect of whisker properties

Fig. 5 shows the change in contact localisation performance as a measure of the operating range W for equivalent T_{1D} and T_{2D} whiskers with length $l_w = 200$ mm, sensor location $l_s = 20$ mm ($0.1l_w$), and momentum factor $w_b \times v_c = 1$ mm^2/s, as the taper deviated from T^*. This shows that the shape of the whisker drives contact localisation performance, with optimally tapered T_{2D} whiskers naturally possessing a larger operating range than equivalent optimised T_{1D} whiskers. A rapid reduction in W was observed for $T < T^*$, which eventually decayed to zero. The rate of decay was greater for the T_{1D} whisker. For $T > T^*$, W^* decreased gradually towards an asymptote. This rate of decay was equal for both T_{1D} and T_{2D} whiskers. Essentially, deviation of the whisker taper from the optimal was detrimental to the whisker performance, but increasing the whisker taper above the optimal (i.e., more taper) was less detrimental than decreasing the taper from optimal.

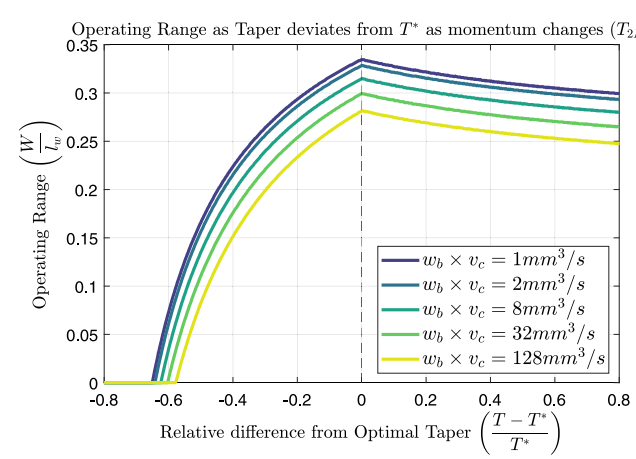
Fig. 6 shows the change in whisker performance as a measure of W as a result of deviating from T^* for T_{2D} whiskers across varying whisker lengths (Fig. 6(a)), varying sensor locations (Fig. 6(b)), and varying momentum factors (Fig. 6(c)). Refer to Fig. E.1 in Appendix E for this relationship for T_{1D} tapers.



(a) Operating range change for varying whisker length (T_{2D}).



(b) Operating range change for varying sensor location (T_{2D}).



(c) Operating range change for varying whisker length (T_{2D}).

Fig. 6. Operating range width for T_{2D} whiskers as the taper deviates from optimal for **(a)** varying whisker length ($l_s = 0.1l_w$, $w_b \times v_c = 1$), **(b)** varying sensor location ($l_w = 200$ mm, $w_b \times v_c = 1$), and **(c)** varying momentum factor ($l_w = 200$ mm, $l_s = 0.1l_w$).

Longer whiskers, whiskers with sensors placed further from the base, and those with less momentum experienced a faster reduction in performance than the contrary as the taper was reduced from the

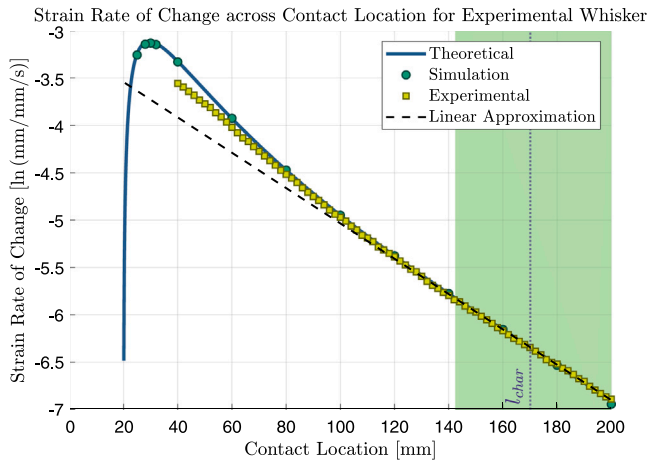


Fig. 7. Strain rate of change across contact location for the experimental whisker. The solid line represents the theoretical strain rate of change curve for the whisker calculated using Eq. (8b), circle datapoints show the results from the simulations, and square datapoints show the experimental results measured using the analysis window. The dotted vertical line at $x = 170$ mm represents the location of the whisker's characteristic inflection point. The shaded vertical area (143 mm $leq\ x \leq 200$ mm) represents the linear operating range of the whisker based on Eq. (11). The dashed line represents the linear approximation of the strain rate of change centred at the inflection point.

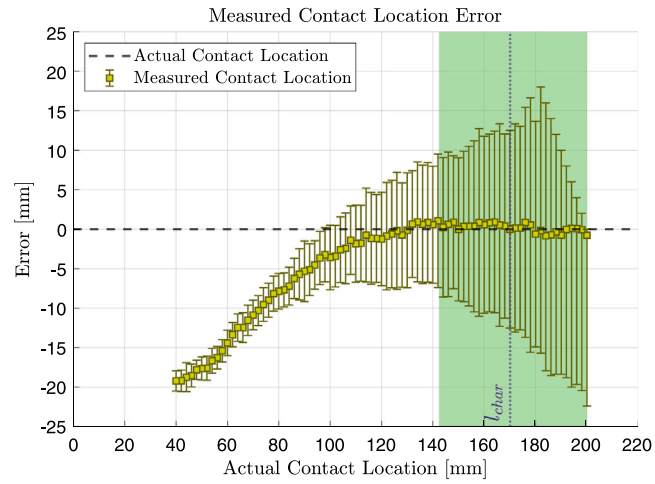


Fig. 8. Error between the measured contact location using the inverse linear approximation model and the actual contact location across the whisker length. Data points represent the average value, and error bars were $\pm 2\sigma$ of the measurements within the analysis window (0.1 s). The dotted vertical line at $x = 170$ mm represents the location of the whisker's characteristic inflection point. The shaded vertical area (143 mm $leq\ x \leq 200$ mm) represents the linear operating range of the whisker based on Eq. (11).

optimal ($T < T^*$). The rate of decay was equivalent for T_{1D} and T_{2D} whiskers as taper increased from optimal ($T > T^*$), and did not change across whisker length, sensor location or momentum factor. The whisker performance was superior for longer whiskers and for those with minimal momentum. Whiskers with sensors placed further towards the tip also showed superior performance, except for cases in which the relative difference from the optimal taper fell below −0.52 for which the inverse was true.

Simulation results for whiskers made from different materials, but having the same geometry, showed that there was no difference in the strain rate of change of the whisker for a given translational deflection rate of change at equivalent contact locations.

3.3. Performance of an electromechanical whisker with optimal taper

To demonstrate the utility of this approach, parameters that were practically useful for a small ground robot operating in confined, cluttered spaces were selected. The parameters used were a length of l_w=200 mm to enable object avoidance without reliance on LiDAR, sensor location of l_s=20 mm to increase the total operating range, base width of w_b=4.5 mm for maximum achievable flexibility based on the strain gauge dimensions, and a contact speed of v_c=16 mm/s to maximise the strain rate of change response with the motorised linear rail. Based on Eq. (10b), T^*_{2D}=8.84 for this whisker system. From Eqs. (11) and (12), the size of operating range was approximately 58 mm ($0.283l_w$), ranging between 143 mm $\leq l_c \leq$ 200 mm, with the characteristic inflection point located at 173 mm ($0.864l_w$).

Following 3D printing, the experimental whisker used was measured to have a taper of T_{2D}=9.0 (w_b=4.5 mm and w_t=1.5 mm), i.e., slightly larger than the theoretical optimal. However, the printed whisker had an estimated operating range of only 0.1 mm ($0.001l_w$) smaller than optimal. The characteristic inflection point was located at l_{char}=170 mm, a 3 mm shift towards the base compared to the theoretical optimal.

The specific equation for the strain rate of change due to contact location for the experimental whisker was therefore given by Eq. (13):

$$\dot{\epsilon}_{s_{T_{2D}}}(l_c) = \frac{0.664(l_c - 20)(200 - 0.67l_c)}{l_c^3}, \quad \{20 \text{ mm} \leq l_c \leq 200 \text{ mm}\} \quad (13)$$

The specific linear approximation for the experimental whisker was given by Eq. (14):

$$\hat{\dot{\epsilon}}_{s_{T_{2D}}}(l_c) = \exp\left(-0.0187l_c - 3.17\right), \quad \{143 \text{ mm} \leq l_c \leq 200 \text{ mm}\} \quad (14)$$

Fig. 7 compares the measured rate of change of the whisker strain across contact location from the experimental whisker to that from simulations, theoretical calculations and the derived linear approximation.

The simulation results showed an almost perfect correlation with the theoretical model, returning an average R^2 value of 0.99 and RMSE of 0.00034. This indicated that the derived theoretical model, and therefore the ascertained effect of the whisker's properties as described in Section 3.2, hold true.

The experimental results followed the shape of the theoretical strain rate of change and contact location relationship closely for the majority of the whisker length, returning an R^2 value of 0.95 and RMSE of 0.0017 for all tested contact locations. Within the linear operating range, the R^2 value improved to 0.998 with RMSE of 0.000025, whereas outside of the linear operating range the R^2 value was 0.92 with RMSE of 0.0021. This indicated that the relationship between strain rate of change and contact location could be accurately predicted within the specified operating range when using the linear approximation model, with increasing error as the contact location moved further outside of the operating range.

Fig. 8 shows the difference between the measured contact location using the inverse of the linear approximation model and the actual location of the contact during the experiment, i.e., the error. Data points represent the measured contact location using the analysis window, and error bars represent two standard deviations of the measurements using the low-pass filter. Contact location measurements were clamped at a maximum of 200 mm (length of the whisker). As a result of this clamping, the error for some contacts made near to the whisker tip was asymmetrical as the positive error was restricted to the difference between the prediction using the analysis window and the whisker length. The difference between the actual contact location and the measured contact location within the analysis window was 0.19 ± 0.55 mm for contacts within the linear operating range. This was equivalent to 0.095% error, matching the defined 0.1% error limit for the operating range. The absolute error increased with an increasing rate of change as the contact location moved out of the linear operating range. For this range, the average error and standard deviation was −7.17 ± 6.65 mm.

The largest error occurred at the lowest contact location of 40 mm where the error was −19.2 mm. The error within the analysis window decreased as the contact location moved closer to the whisker base due to the increasing rate of change of the strain rate of change as the contact location approached the base.

4. Discussion

This research shows that a sufficient magnitude of taper is required for accurate 2D contact localisation to be achievable using a linear approximation of the whisker's strain rate of change. This is the smallest taper for which the size of the operating range (W) is greater than zero, which is always greater than 1 (meaning the whisker has a smaller tip than base, i.e., it is tapered) for all whiskers of realistic properties. This can be observed as the x-intercept in the examples given in Fig. 6. This follows the findings of [40] which showed that contact localisation is more accurate for tapered whiskers than untapered whiskers. This is despite the different sensing method used in this work, which points towards tapered whiskers universally providing greater accuracy than untapered whiskers. This was also observed in [30], where untapered whiskers returned lower contact localisation accuracy than tapered whiskers once the whisker deflection increased beyond 5°. The shape of the whisker was also noted to be of importance for contact localisation accuracy, with T_{2D} whiskers performing better than T_{1D} whiskers of equivalent specifications, as observed in Fig. 5.

Our results expand on this by quantifying the taper magnitude that provides optimal performance (T^*) defined by having the widest operating range including the whisker tip through Eq. (10). In realistic scenarios (e.g., whisker lengths below 10s of metres, base widths in the order of mm and contact speeds in the order of mm/s to m/s), T^* is always smaller than infinity (i.e., the whisker never tapers to a point where w_t=0), indicating that whiskers are best modelled as truncated prisms. This is also seen as the performance decays as $T \rightarrow +\infty$ (see Fig. 6). This is in agreement with the common approach of approximating rat whiskers as truncated cones [53,54]. However, as W does not approach 0 for any $T > T^*$, whiskers tapered to a point are still practical solutions, albeit sub-optimal when using our linearised signal processing approach.

Thicker whiskers require smaller tapers than thinner whiskers for optimal performance according to Eq. (10). This correlates with the geometric properties of real rat whiskers, where the longer, thicker, caudal whiskers are tapered less than the shorter, thinner, rostral whiskers [21].

An optimally tapered whisker's point of most accuracy (the characteristic inflection point), l^*_{char}, was found to move towards the whisker tip as the whisker's momentum increases (see Eq. (12)). As the momentum is proportional to w_b and v_c, this implies that for a whisker of given physical properties, the point at which contact locations can be most accurately discriminated shifts with the whisk speed. This quantifies the previously suggested "sweet spot" between whisk speed and contact location in [52]. From a practical perspective, this implies that for applications where the contact location can be controlled, contacts should be made at the whisker's characteristic inflection point, which requires a constant, appropriate whisk speed. This is reflective of rat behaviour, where they are observed to favour contacts near or at the whisker tip during exploration [24].

Whiskers of a given length, base width, and sensor location perform optimally at similar taper magnitudes regardless of whether or not they possess a biologically mimetic medulla. However, considering two optimally-tapered whiskers, one with a medulla and one without, the lower mass of the whisker possessing the medulla reduces sagging under gravity, the energy required to perform whisking, and the bending stiffness [20]. This ensures correlation with the derived strain rate of change to contact location relationship, reduces power consumption, and enhances mobility in confined spaces. However, due to the lowered stiffness at the base, the resolution of the whisker would decrease [20]. Therefore, in robotic mapping scenarios, it would be optimal for electromechanical whiskers to possess a medulla as long as the required sensitivity is achievable and the manufacturing process was sufficiently precise so as to not compromise the structural integrity of the whisker.

5. Conclusion

This paper quantified the optimal balance between the geometry, design and control of an electromechanical whisker for achieving accurate and efficient localisation of radial, translational contacts, along with their distinct effects. The use of a whisker's optimal taper (dependent upon whisker's length, base width, distance of the sensor from the base, and contacting speed, but independent from the presence of a biologically mimetic medulla and the whiskers material) enables the largest region of the whisker length, including the tip, to be capable of accurately localising contacts. The use of an optimally tapered whisker based on this can enhance whisker-based robotic mapping applications in terms of object localisation accuracy, whilst utilising the whisker's entire length. Whiskers tapered in 2D are able to accurately localise contacts over a larger portion of the whisker than those tapered in 1D. This was also true for longer whiskers and those with less basal momentum (i.e., smaller cross-sections and/or slower whisk speeds). Additionally, whiskers with the strain sensing element positioned further from the base showed better performance in most cases. The developed theorem returns a characteristic contact location that results in zero error for a whisker, which is also dependent on the aforementioned whisker characteristics. This can be useful in applications requiring a controlled contact location.

These theorems are only valid for contacts occurring in a single plane and for translational whisker motion, and future work is required to quantify the relationship of whisker geometry on performance for both 3D contacts and for angular whisking behaviour. Only medullae of proportions fixed to mimic the proportions of rat whisker medullae were explored, and the impact of medullae of different proportions and shapes on the whisker performance should be analysed as this is potential grounds for further optimisation of electromechanical whisker design. The impact of the whisker dynamics on the contact localisation performance has not been explored in this research, with the whisker contact treated under static conditions. It is expected that, during translational whisker movement additional oscillatory noise would arise in the strain signal, which would reduce the precision of contact localisation. We can hypothesise that increasing the damping parameter of the whisker (e.g., by making it thicker or manufacturing it from a stiffer material) or increasing the time constant of the low-pass filter would account for this at the cost of flexibility and time to discriminate contacts, but the quantification of this should be grounds for future work.

The ensuing goal is to extend this research to the domain of measuring an object's movability through the use of whiskers of here-derived optimal tapers by using contacts located at the characteristic contact location.

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CRediT authorship contribution statement

Simon P. Pegoli: Writing – original draft, Visualization, Validation, Software, Resources, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Phillip S.M. Skelton:** Writing – review & editing, Validation, Supervision, Methodology, Conceptualization. **Russell S.A. Brinkworth:** Writing – review & editing, Validation, Supervision, Resources, Project administration, Methodology, Funding acquisition, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

Appendix A

A.1. Derivation of scale of diversion of second moment of area for T_{1D} whiskers possessing a medulla

For a T_{1D} whisker with base width w_b and length l_w possessing a medulla of square basal cross-section with width w_m, the second moment of area at the whisker base (I_{m_b}) is:

$$I_{m_b} = \frac{{w_b}^4 - {w_m}^4}{12} \tag{A.1}$$

Substituting Eq. (9a) into Eq. (A.1), I_{m_b} becomes:

$$I_{m_b} = \frac{{w_b}^4 - (0.324 w_b)^4}{12} \tag{A.2}$$

As the height of the whisker and the height of the medulla remain constant throughout its length, the second moment of area at the sensor location l_s (I_{m_s}) is:

$$I_{m_s} = \frac{w_b {w_s}^3 - w_m {w_{m_s}}^3}{12} \tag{A.3}$$

Substituting Eqs. (3a), (9a) and (9c) into Eq. (A.3), I_{m_s} becomes:

$$I_{m_s} = \frac{w_b\left(w_b\left(1+\frac{l_s}{l_w}(1/T-1)\right)\right)^3 - 0.324 w_b\left(w_b\left(0.324-0.415\frac{l_s}{l_w}\right)\right)^3}{12} \tag{A.4}$$

The scale of diversion of the second moment of area at the sensor location from that at the whisker base is therefore:

$$\nabla I_m = \frac{I_{m_s}}{I_{m_b}} = 1.011\left(\left(1+\frac{l_s}{l_w}(1/T-1)\right)^3 - 0.324\left(0.324-0.415\frac{l_s}{l_w}\right)^3\right) \tag{A.5}$$

Taking the ratio of ∇I_m to $\nabla I_{T_{1D}}$ (from Eq. (4a)):

$$\frac{\nabla I_m}{\nabla I_{T_{1D}}} = 1.011 - \frac{0.324\left(0.324-0.415\frac{l_s}{l_w}\right)^3}{\left(1+\frac{l_s}{l_w}(1/T-1)\right)^3}, \qquad \begin{array}{c} T \geq 1 \\ 0 \leq l_s/l_w \leq 0.125 \end{array} \tag{A.6}$$

Substituting $T = \infty$ and $ls/lw = 0$ into Eq. (A.6) provides the lower limit of $\frac{\nabla I_m}{\nabla I_{T_{1D}}}$:

$$\min \frac{\nabla I_m}{\nabla I_{T_{1D}}} = 1.011 - \frac{0.324\,(0.324-0.415\times 0)^3}{(1+0(1/\infty-1))^3} = 1.011 - 0.324^4 = 0.99 \tag{A.7}$$

Substituting $T = 1$ and $ls/lw = 0.125$ provides the upper limit of $\frac{\nabla I_m}{\nabla I_{T_{1D}}}$:

$$\max \frac{\nabla I_m}{\nabla I_{T_{1D}}} = 1.011 - \frac{0.324\,(0.324-0.415\times 0.125)^3}{(1+0.125(1/1-1))^3} = 1.004 \tag{A.8}$$

Therefore, $0.99 \leq \frac{\nabla I_m}{\nabla I_{T_{1D}}} \leq 1.004$, so $\nabla I_m \approxeq \nabla I_{T_{1D}}$.

A.2. Derivation of scale of diversion of second moment of area for T_{2D} whiskers possessing a medulla

For a T_{2D} whisker with base width w_b and length l_w possessing a medulla of circular basal cross-section with width w_m, the second moment of area at the whisker base (I_{m_b}) is:

$$I_{m_b} = \frac{\pi}{4}\left(\left(\frac{w_b}{2}\right)^4 - \left(\frac{w_m}{2}\right)^4\right) \tag{A.9}$$

Substituting Eq. (9a) into Eq. (A.9), I_{m_b} becomes:

$$I_{m_b} = \frac{\pi}{4}\left(\left(\frac{w_b}{2}\right)^4 - \left(\frac{0.324 w_b}{2}\right)^4\right) \tag{A.10}$$

As both the height of the whisker and the height of the medulla taper linearly throughout its length (at different rates), the second moment of area at the sensor location l_s (I_{m_s}) is:

$$I_{m_s} = \frac{\pi}{4}\left(\left(\frac{w_s}{2}\right)^4 - \left(\frac{w_{m_s}}{2}\right)^4\right) \tag{A.11}$$

Substituting Eqs. (3b), (9a) and (9c) into Eq. (A.11), I_{m_s} becomes:

$$I_{m_s} = \frac{\pi}{4}\left(\left(\frac{w_b\left(1+\frac{l_s}{l_w}\left(\sqrt{1/T}-1\right)\right)}{2}\right)^4 - \left(\frac{w_b\left(0.324-0.415\frac{l_s}{l_w}\right)}{2}\right)^4\right) \tag{A.12}$$

The scale of diversion of the second moment of area at the sensor location from that at the whisker base is therefore:

$$\nabla I_m = \frac{I_{m_s}}{I_{m_b}} = 1.011\left(\left(1+\frac{l_s}{l_w}\left(\sqrt{1/T}-1\right)\right)^4 - \left(0.324-0.415\frac{l_s}{l_w}\right)^4\right) \tag{A.13}$$

Taking the ratio of ∇I_m to $\nabla I_{T_{2D}}$ (from Eq. (4b)):

$$\frac{\nabla I_m}{\nabla I_{T_{2D}}} = 1.011 - \frac{\left(0.324-0.415\frac{l_s}{l_w}\right)^4}{\left(1+\frac{l_s}{l_w}(1/T-1)\right)^4}, \qquad \begin{array}{c} T \geq 1 \\ 0 \leq l_s/l_w \leq 0.125 \end{array} \tag{A.14}$$

Substituting $T = \infty$ and $ls/lw = 0$ into Eq. (A.14) provides the lower limit of $\frac{\nabla I_m}{\nabla I_{T_{2D}}}$:

$$\min \frac{\nabla I_m}{\nabla I_{T_{2D}}} = 1.011 - \frac{(0.324-0.415\times 0)^4}{(1+0(1/\infty-1))^4} = 1.011 - 0.324^4 = 0.99 \tag{A.15}$$

Substituting $T = 1$ and $ls/lw = 0.125$ provides the upper limit of $\frac{\nabla I_m}{\nabla I_{T_{2D}}}$:

$$\max \frac{\nabla I_m}{\nabla I_{T_{2D}}} = 1.011 - \frac{(0.324-0.415\times 0.125)^4}{(1+0.125(1/1-1))^4} = 1.006 \tag{A.16}$$

Therefore, $0.99 \leq \frac{\nabla I_m}{\nabla I_{T_{2D}}} \leq 1.006$, so $\nabla I_m \approxeq \nabla I_{T_{2D}}$.

Appendix B

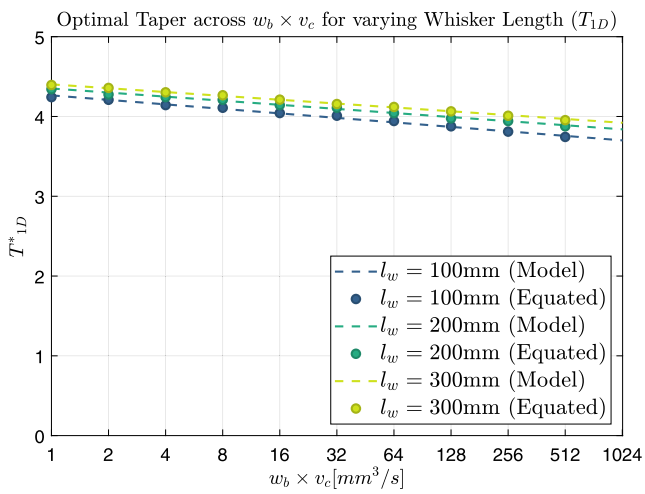
See Figs. B.1 and B.2.

Appendix C

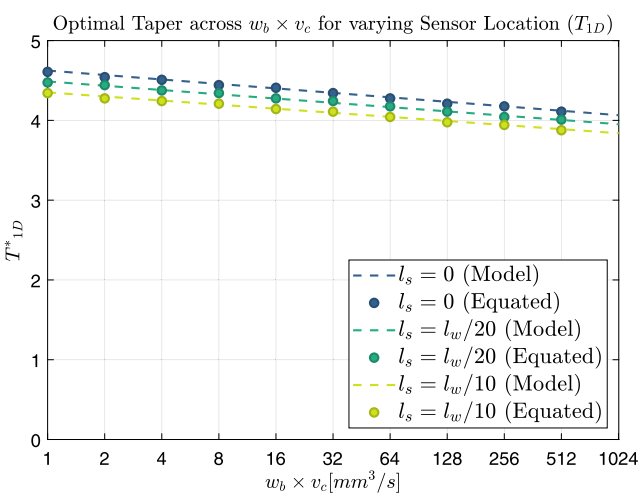
See Figs. C.1 and C.2.

Appendix D

See Figs. D.1 and D.2.

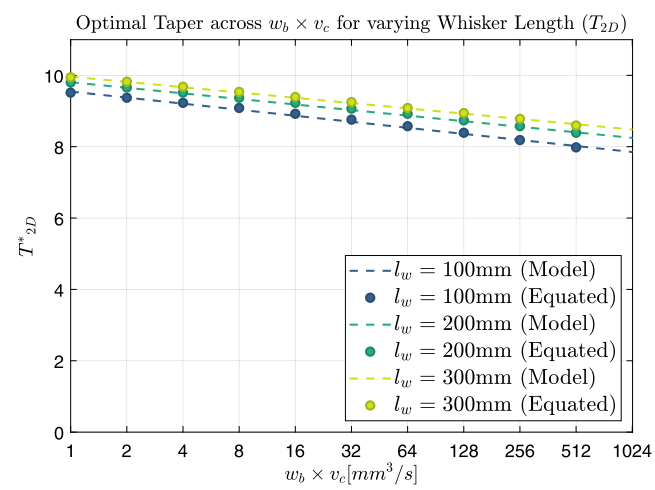


(a) Varying whisker length.

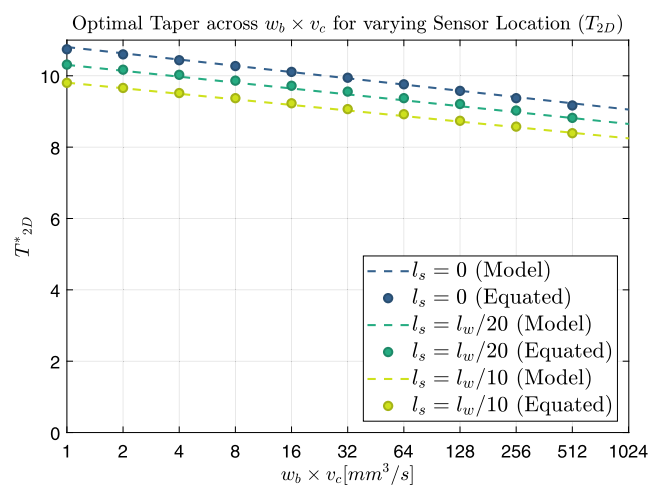


(b) Varying sensor location.

Fig. B.1. Optimal taper magnitude for T_{1D} whiskers across the momentum factor for **(a)** varying whisker length ($l_s = l_w/10$), and **(b)** varying sensor location (l_w = 200 mm). Datapoints show the optimal taper calculated directly from the theory for the respectively labelled parameter values. Dashed lines show the line of best fit from Eq. (10a) for the respectively labelled parameter values.

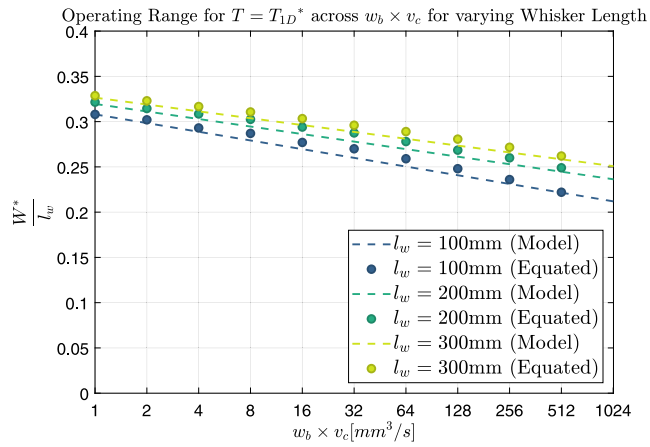


(a) Varying whisker length.

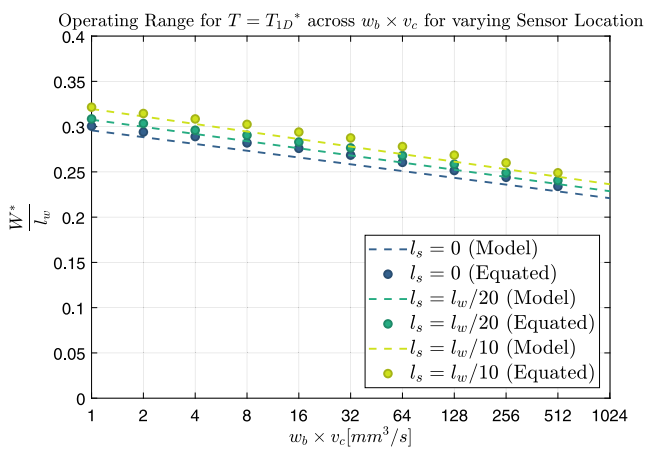


(b) Varying sensor location.

Fig. B.2. Optimal taper magnitude for T_{2D} whiskers across the momentum factor for **(a)** varying whisker length ($l_s = l_w/10$), and **(b)** varying sensor location (l_w = 200 mm). Datapoints show the optimal taper calculated directly from the theory for the respectively labelled parameter values. Dashed lines show the line of best fit from Eq. (10b) for the respectively labelled parameter values.

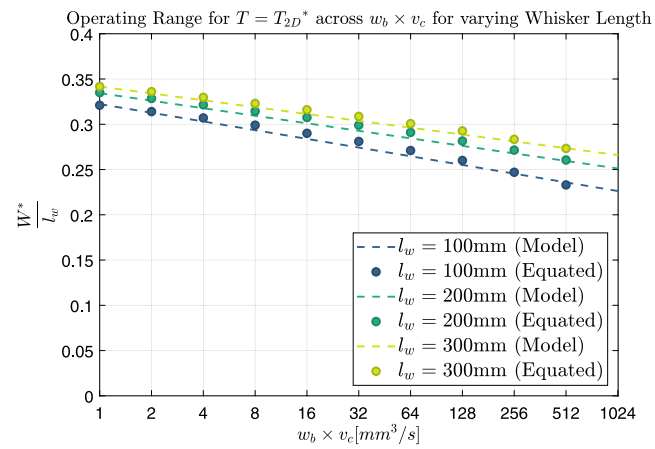


(a) Varying whisker length.

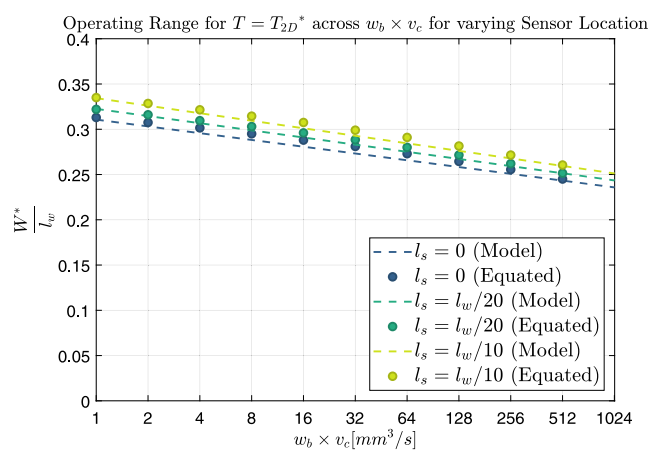


(b) Varying sensor location.

Fig. C.1. Size of the linear operating range for T_{1D} whiskers with optimal taper across the momentum factor for **(a)** varying whisker length ($l_s = l_w/10$), and **(b)** varying sensor location (l_w = 200 mm). Datapoints show the linear operating range size calculated directly from the theory for the respectively labelled parameter values. Dashed lines show the line of best fit from Eq. (11) for the respectively labelled parameter values.

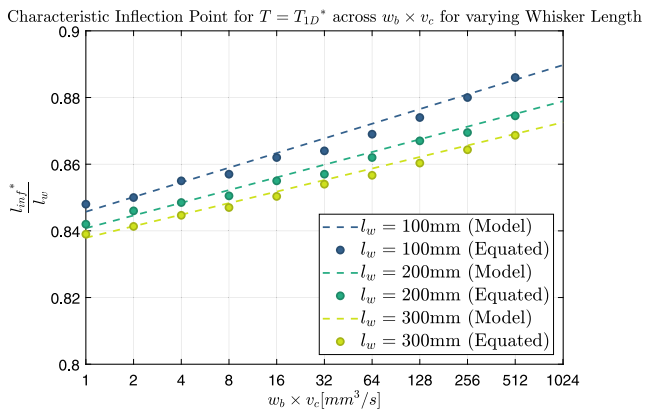


(a) Varying whisker length.

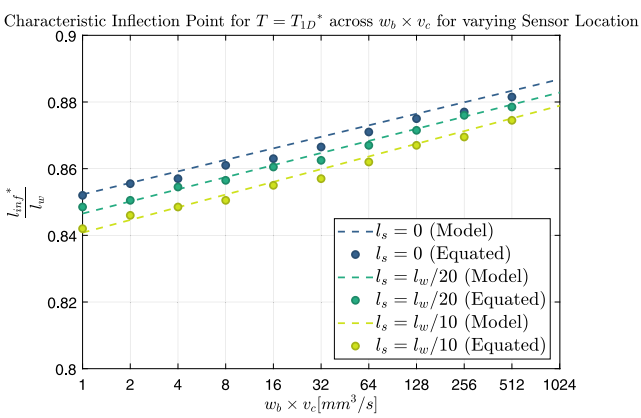


(b) Varying sensor location.

Fig. C.2. Size of the linear operating range for T_{2D} whiskers with optimal taper across the momentum factor for **(a)** varying whisker length ($l_s = l_w/10$), and **(b)** varying sensor location (l_w = 200 mm). Datapoints show the linear operating range size calculated directly from the theory for the respectively labelled parameter values. Dashed lines show the line of best fit from Eq. (11) for the respectively labelled parameter values.

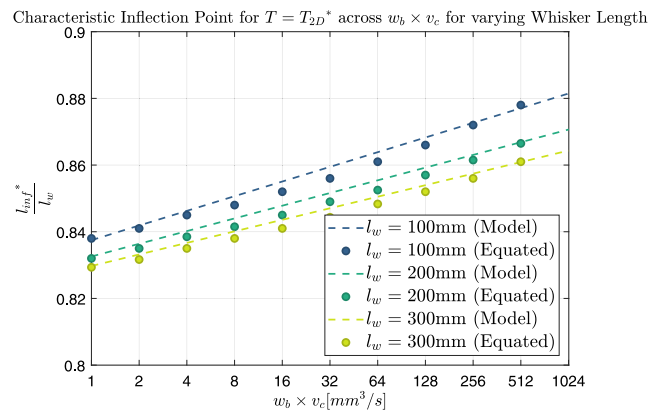


(a) Varying whisker length.

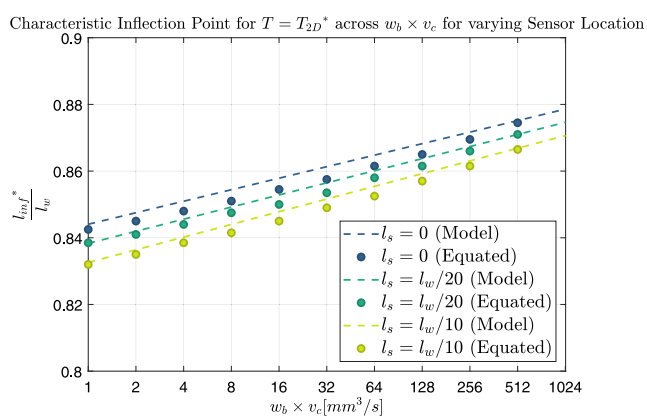


(b) Varying sensor location.

Fig. D.1. Location of the characteristic inflection point as a ratio of the whisker length for T_{1D} whiskers with optimal taper across the momentum factor for **(a)** varying whisker length ($l_s = l_w/10$), and **(b)** varying sensor location ($l_w = 200$ mm). Datapoints show the characteristic inflection point location calculated directly from the theory for the respectively labelled parameter values. Dashed lines show the line of best fit from Eq. (12) for the respectively labelled parameter values.

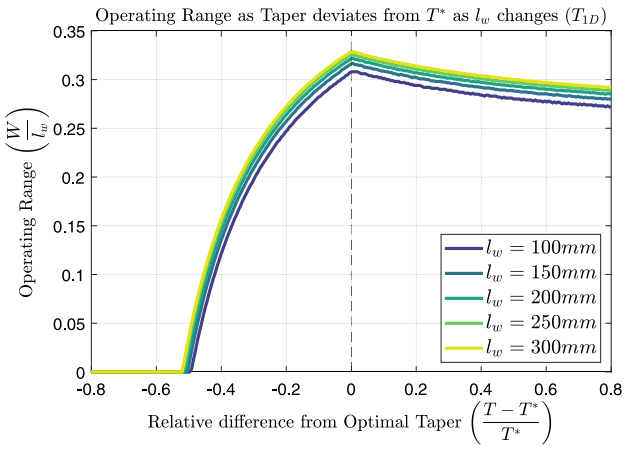


(a) Varying whisker length.

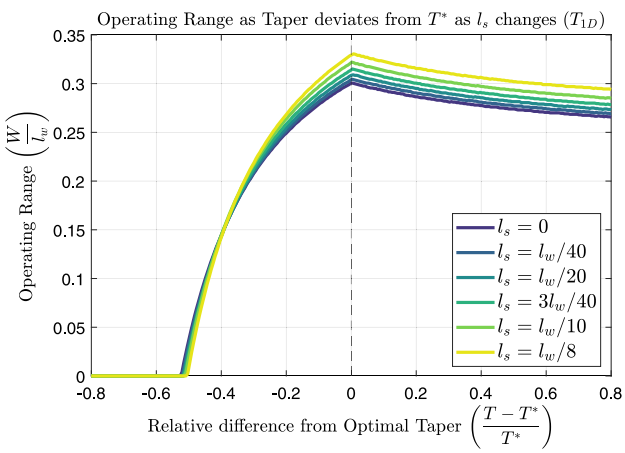


(b) Varying sensor location.

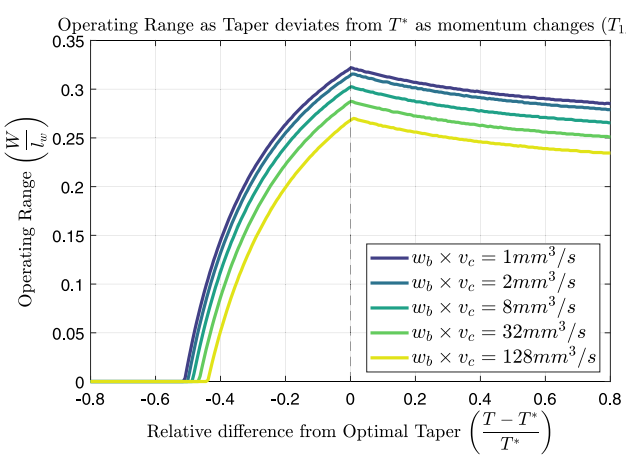
Fig. D.2. Location of the characteristic inflection point as a ratio of the whisker length for T_{2D} whiskers with optimal taper across the momentum factor for **(a)** varying whisker length ($l_s = l_w/10$), and **(b)** varying sensor location ($l_w = 200$ mm). Datapoints show the characteristic inflection point location calculated directly from the theory for the respectively labelled parameter values. Dashed lines show the line of best fit from Eq. (12) for the respectively labelled parameter values.



(a) Operating range change for varying whisker length (T_{1D}).



(b) Operating range change for varying sensor location (T_{1D}).



(c) Operating range change for varying whisker length (T_{1D}).

Fig. E.1. Operating range size for T_{1D} whiskers as the taper deviates from optimal for **(a)** varying whisker length ($l_s = 0.1l_w$, $w_b \times v_c = 1$), **(b)** varying sensor location ($l_w = 200$ mm, $w_b \times v_c = 1$), and **(c)** varying momentum factor ($l_w = 200$ mm, $l_s = 0.1l_w$).

Appendix E

See Fig. E.1.

References

[1] T. Buratowski, J. Garus, M. Giergiel, A. Kudriashov, Real-time 3D mapping in isolated industrial terrain with use of mobile robotic vehicle, Electronics 11 (2022) 2086, http://dx.doi.org/10.3390/electronics11132086.

[2] J.P. Queralta, J. Taipalmaa, B.C. Pullinen, V.K. Sarker, T.N. Gia, H. Tenhunen, M. Gabbouj, J. Raitoharju, T. Westerlund, Collaborative multi-robot search and rescue: Planning, coordination, perception, and active vision, IEEE Access 8 (2020) 191617–191643, http://dx.doi.org/10.1109/ACCESS.2020.3030190.

[3] G.S. Hornby, S. Takamura, J. Yokono, O. Hanagata, T. Yamamoto, M. Fujita, Evolving robust gaits with AIBO, in: Proceedings Proceedings 2000 ICRA, San Francisco, CA, USA, 2000, http://dx.doi.org/10.1109/ROBOT.2000.846489.

[4] J.L. Jones, Robots at the tipping point: the road to iRobot Roomba, IEEE Robot. Autom. Mag. 13 (2006) 76–78, http://dx.doi.org/10.1109/MRA.2006.1598056.

[5] H. Durrant-Whyte, T. Bailey, Simultaneous localization and mapping: Part I, IEEE Robot. Autom. Mag. 13 (2006) 99–110, http://dx.doi.org/10.1109/MRA.2006.1638022.

[6] I. Lluvia, E. Lazkano, A. Ansuategi, Active mapping and robot exploration: A survey, Sensors 21 (2021) 2445, http://dx.doi.org/10.3390/s21072445.

[7] C. Rosales, F. Spinelli, M. Gabiccini, C. Zito, J.L. Wyatt, GPAtlasRRT: A local tactile exploration planner for recovering the shape of novel objects, Int. J. Humanoid Robot. 15 (2018) 1850014-1–1850014-27, http://dx.doi.org/10.1142/S0219843618500147.

[8] K.A. Nichols, A.M. Okamura, Active multiobject exploration and recognition via tactile whiskers, IEEE Trans. Robot. 38 (2022) 3479–3497, http://dx.doi.org/10.1109/TRO.2022.3182487.

[9] S. Jia, L. Zhang, V.J. Santos, Autonomous tactile localization and mapping of objects buried in granular materials, IEEE Robot. Autom. Lett. 7 (2022) 9953–9960, http://dx.doi.org/10.1109/LRA.2022.3191955.

[10] Yue. Y, Y. Tang, Q. Wang, W. Xiao, J. Liu, J. Wang, M. Chen, G. Wu, B. Su, Active perception in non-visual recognition environments by stretchable tentacle sensor arrays, ACS Appl. Mater. Interfaces 14 (2022) 26913–26922, http://dx.doi.org/10.1021/acsami.2c04717.

[11] Y. Mulgaonkar, W. Liu, D. Thakur, K. Daniilidis, C.J. Taylor, V. Kumar, The tiercel: A novel autonomous micro aerial vehicle that can map the environment by flying into obstacles, in: Proceedings of the IEEE International Conference on Robotics and Automation, Paris, France, 31 May–31 August 2020, 2020, http://dx.doi.org/10.1109/ICRA40945.2020.9197269.

[12] M. Bajracharya, A. Howard, L.H. Matthies, B. Tang, M. Turmon, Autonomous off-road navigation with end-to-end learning for the LAGR program, J. Field Robotics 26 (2009) 3–25, http://dx.doi.org/10.1002/rob.20269.

[13] R.A. Grant, V.G.A. Goss, What can whiskers tell us about mammalian evolution, behaviour, and ecology? Mammal Rev. 52 (2022) 148–163, http://dx.doi.org/10.1111/mam.12253.

[14] R.A. Grant, V. Breakell, T.J. Prescott, Whisker touch sensing guides locomotion in small, quadrupedal mammals, Proc. R. Soc. B 285 (2018) 20180592, http://dx.doi.org/10.1098/rspb.2018.0592.

[15] M. Brecht, B. Preilowski, M.M. Merzenich, Functional architecture of the mystacial vibrissae, Behav. Brain Res. 84 (1997) 81–97, http://dx.doi.org/10.1016/S0166-4328(97)83328-1.

[16] T.J. Prescott, B. Mitchinson, R.A. Grant, Vibrissal behavior and function, Scholarpedia 6 (2011) 6642, http://dx.doi.org/10.4249/scholarpedia.6642.

[17] F. Gonçalves, R.S. Bovendorp, G. Beca, C. Bello, R. Costa-Pereira, R.L. Muylaert, R.R. Rodarte, N. Villar, R. Souza, M.E. Graipel, et al., ATLANTIC MAMMAL TRAITS: a data set of morphological traits of mammals in the Atlantic Forest of South America, Ecology (2018) http://dx.doi.org/10.1002/ecy.2106.

[18] D. Voges, K. Carl, G.J. Klauer, R. Uhlig, C. Schilling, C. Behn, H. Witte, Structural characterization of the whisker system of the rat, IEEE Sens. J. 12 (2011) 332–339, http://dx.doi.org/10.1109/JSEN.2011.2161464.

[19] C.M. Williams, E.M. Kramer, The advantages of a tapered whisker, PLOS ONE 5 (2010) e8806, http://dx.doi.org/10.1371/journal.pone.0008806.

[20] A.E. Yang, H.M. Belli, M.J.Z. Hartmann, Quantification of vibrissal mechanical properties across the rat mystacial pad, J. Neurophysiol. 121 (2019) 1879–1895, http://dx.doi.org/10.1152/jn.00869.2016.

[21] H.M. Belli, A.E. Yang, C.S. Bresee, M.J.Z. Hartmann, Variations in vibrissal geometry across the rat mystacial pad: base diameter, medulla, and taper, J. Neurophysiol. 117 (2017) 1807–1820, http://dx.doi.org/10.1152/jn.00054.2016.

[22] L.A. Huet, M.J.Z. Hartmann, The search space of the rat during whisking behavior, J. Exp. Biol. 217 (2014) 3365–3376, http://dx.doi.org/10.1242/jeb.105338.

[23] G.E. Carvell, D.J. Simons, Biometric analyses of vibrissal tactile discrimination in the rat, J. Nuerosci. 10 (1990) 2638–2648, http://dx.doi.org/10.1523/JNEUROSCI.10-08-02638.1990.

[24] B. Mitchinson, C.J. Martin, R.A. Grant, T.J. Prescott, Feedback control in active sensing: rat exploratory whisking is modulated by environmental contact, Proc. R. Soc. B 274 (2007) 1035–1041, http://dx.doi.org/10.1098/rspb.2006.0347.

[25] E. Zucker, W.I. Welker, Coding of somatic sensory input by vibrissae neurons in the rat's trigeminal ganglion, Brain Res. 12 (1969) 138–156, http://dx.doi.org/10.1016/0006-8993(69)90061-4.

[26] M. Szwed, K. Bagdasarian, E. Ahissar, Encoding of vibrissal active touch, Neuron 40 (2003) 621–630, http://dx.doi.org/10.1016/S0896-6273(03)00671-8.

[27] E. Ahissar, P.M. Knutsen, Object localization with whiskers, Biol. Cybernet. 98 (2008) 449–458, http://dx.doi.org/10.1007/s00422-008-0214-4.

[28] L. Pammer, D.H. O'Connor, S.A. Hires, N.G. Clack, D. Huber, E.W. Myers, K. Svoboda, The mechanical variables underlying object localization along the axis of the whisker, J. Neurosci. 33 (2013) 6726–6741, http://dx.doi.org/10.1523/JNEUROSCI.4316-12.2013.

[29] D.B. Polley, J.L. Rickert, R.D. Frostig, Whisker-based discrimination of object orientation determined with a rapid training paradigm, Neurobiol. Learn. Memory 83 (2005) 134–142, http://dx.doi.org/10.1016/j.nlm.2004.10.005.

[30] N.H. Nguyen, T.D. Ngo, D.Q. Nguyen, Contact distance estimation by a soft active whisker sensor based on morphological computation, in: 2020 8th IEEE RAS/EMBS International Conference for Biomedical Robotics and Biomechatronics (BioRob), New York, NY, USA, 2020, http://dx.doi.org/10.1109/BioRob49111.2020.9224314.

[31] J.H. Solomon, M.J. Hartmann, Robotic whiskers used to sense features, Nature 443 (2006) 525, http://dx.doi.org/10.1038/443525a.

[32] C. Xiao, S. Xu, W. Wu, J. Wachs, Active multiobject exploration and recognition via tactile whiskers, IEEE Trans. Robot. 38 (2022) 3479–3497, http://dx.doi.org/10.1109/TRO.2022.3182487.

[33] W. Deer, P.E.I. Pounds, Lightweight whiskers for contact, pre-contact, and fluid velocity sensing, IEEE Robot. Autom. Lett. 4 (2019) 1978–1984, http://dx.doi.org/10.1109/LRA.2019.2899215.

[34] Y. Zhang, S. Yan, Z. Wei, X. Chen, T. Fukuda, Q. Shi, A small-scale, rat-inspired whisker sensor for the perception of a biomimetic robot: Design, fabrication, modeling, and experimental characterization, IEEE Robot. Autom. Mag. 29 (2022) 115–126, http://dx.doi.org/10.1109/MRA.2022.3182870.

[35] C. Zhao, Q. Jiang, Y. Li, A novel biomimetic whisker technology based on fiber Bragg grating and its application, Meas. Sci. Technol. 28 (2017) 095104, http://dx.doi.org/10.1088/1361-6501/aa7d36.

[36] M.J. Pearson, B. Mitchinson, J. Welsby, T. Pipe, T.J. Prescott, Scratchbot: Active tactile sensing in a whiskered mobile robot, in: Proceedings of the International Conference on Simulation of Adaptive Behavior, Paris, 25-28 August 2010, 2010, http://dx.doi.org/10.1007/978-3-642-15193-4_9.

[37] M.A. Lin, E. Reyes, J. Bohg, M.R. Cutkosky, Whisker-inspired tactile sensing for contact localization on robot manipulators, in: Proceedings of the IEEE/RSJ International Conference on Intelligent Robots and Systems, Kyoto, Japan, 23-27 October 2022, 2022, http://dx.doi.org/10.1109/IROS47612.2022.9982122.

[38] M.A. Sayegh, H. Daraghma, S. Mekid, S. Bashmal, Review of recent bio-inspired design and manufacturing of whisker tactile sensors, Sensors 22 (2022) 2705, http://dx.doi.org/10.3390/s22072705.

[39] M. Evans, C.W. Fox, M.J. Pearson, T.J. Prescott, Spectral template based classification of robotic whisker sensor signals in a floor texture discrimination task, in: Proceedings TAROS, 2009, pp. 19–24.

[40] D.W. Collinson, H.M. Emnett, J. Ning, M.J.Z. Hartmann, L.C. Brinson, Tapered polymer whiskers to enable three-dimensional tactile feature extraction, Soft Robot. 8 (2021) 44–58, http://dx.doi.org/10.1089/soro.2019.0055.

[41] S. Mekid, H. Daraghma, Experimental ultrasonic sub-surface consolidation of fiber bragg grating for sensorial materials, J. Mater. Process. Technol. 252 (2018) 673–679, http://dx.doi.org/10.1016/j.jmatprotec.2017.10.036.

[42] S. Mekid, H. Daraghma, Sensing signal assessment in sensorial materials: Key embedding conditions, in: Proceedings of International Multi-Conference on Systems, Signals & Devices, Monastir, Tunisia, 20-23 July 2020, 2020, http://dx.doi.org/10.1109/SSD49366.2020.9364207.

[43] C. Fox, M. Evans, M. Pearson, T.J. Prescott, Tactile SLAM with a biomimetic whiskered robot, in: Proceedings of IEEE International Conference on Robotics and Automation, Saint Paul, Minnesota, USA, 14-18 May 2012, 2012, http://dx.doi.org/10.1109/ICRA.2012.6224813.

[44] M.J. Pearson, C. Fox, J.C. Sullivan, T.J. Prescott, T. Pipe, B. Mitchinson, Simultaneous localisation and mapping on a multi-degree of freedom biomimetic whiskered robot, in: Proceedings of the IEEE International Conference on Robotics and Automation, Karlsruhe, Germany, 6-10 May 2013, 2013, http://dx.doi.org/10.1109/ICRA.2013.6630633.

[45] O. Struckmeier, K. Tiwari, M. Salman, M.J. Pearson, V. Kyrki, Vita-slam: A bio-inspired visuo-tactile slam for navigation while interacting with aliased environments, in: Proceedings of the IEEE International Conference on Cyborg and Bionic Systems, Munich, Germany, 19-20 September 2019, 2019, http://dx.doi.org/10.1109/CBS46900.2019.9114526.

[46] C.W. Fox, B. Mitchinson, M.J. Pearson, A.G. Pipe, T.J. Prescott, Contact type dependency of texture classification in a whiskered mobile robot, Auton. Robots 26 (2009) 223–239, http://dx.doi.org/10.1007/s10514-009-9109-z.

[47] N.F. Lepora, M. Evans, C.W. Fox, M.E. Diamond, K. Gurney, T.J. Prescott, Naive Bayes texture classification applied to whisker data from a moving robot, in: Proceedings of the International Joint Conference on Neural Networks, Barcelona, Spain, 18-23 July 2010, 2010, http://dx.doi.org/10.1109/IJCNN.2010.5596360.

[48] H. Wegiriya, N. Herzig, S.A. Abad, S.M.H. Sadati, T. Nanayakkara, A stiffness controllable multimodal whisker sensor follicle for texture comparison, IEEE Sens. J. 20 (2019) 2320–2328, http://dx.doi.org/10.1109/JSEN.2019.2951755.

[49] O.A. Bauchau, J.I.B. Craig, Structural Analysis: With Applications To Aerospace Structures, first ed., Springer Dordrecht, Berlin, Germany, 2009, pp. 173–221, http://dx.doi.org/10.1007/978-90-481-2516-6.

[50] S.A. Hires, L. Pammer, K. Svoboda, D. Golomb, Tapered whiskers are required for active tactile sensation, eLIFE 2 (2013) e01350, http://dx.doi.org/10.7554/eLife.01350.

[51] L.A. Huet, J.W. Rudnicki, M.J.Z. Hartmann, Tactile sensing with whiskers of various shapes: Determining the three-dimensional location of object contact based on mechanical signals at the whisker base, Soft Robot. 4 (2017) 88–102, http://dx.doi.org/10.1089/soro.2016.0028.

[52] M.H. Evans, C.W. Fox, M.J. Pearson, T.J. Prescott, Tactile discrimination using template classifiers: Towards a model of feature extraction in mammalian vibrissal systems, in: Proceedings of the International Conference on Simulation of Adaptive Behavior, Paris, France, 25-28 August 2010, 2010, http://dx.doi.org/10.1007/978-3-642-15193-4_17.

[53] M.J. Hartmann, N.J. Johnson, R.B. Towal, C. Assad, Mechanical characteristics of rat vibrissae: resonant frequencies and damping in isolated whiskers and in the awake behaving animal, J. Neurosci. 23 (2003) 6510–6519, http://dx.doi.org/10.1523/JNEUROSCI.23-16-06510.2003.

[54] J.A. Birdwell, J.H. Solomon, M. Thajchayapong, M.A. Taylor, M. Cheely, R.B. Towal, J. Conradt, M.J.Z. Hartmann, Biomechanical models for radial distance determination by the rat vibrissal system, J. Neurophysiol. 98 (2007) 2439–2455, http://dx.doi.org/10.1152/jn.00707.2006.

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